

Part: **3.2 – MECHANICALLY FASTENED JOINTS**

Chapter: **3240**

Title: **TENSION FITTINGS AND CLIPS**

Revision: **0.3**

Date: **27/07/2021**

1 INTRODUCTION

1.1 METHOD-SUMMARY

This chapter deals with methods of estimating the ultimate load capacity of angles and club-foot fittings that are assembled so as to provide a tensile load path across a join or through an intervening member.

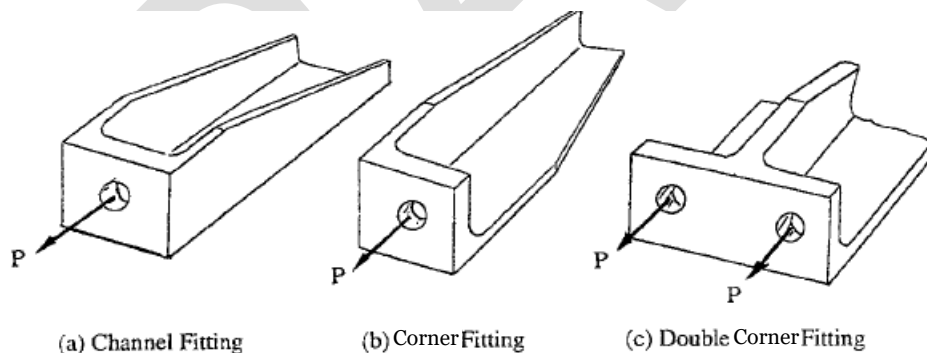


Figure 1. *Typically highly loaded tension fittings (Niu's Fig. 9.9.1)*

For even the simplest joint the complex behaviour under load, resulting from the interplay of stiffnesses of the various elements, is not amenable to accurate analysis. Accordingly, it is only possible to provide simplified models that illustrate the major effects of the many factors that influence joint strength.

Previous paragraphs are extracts from Ref. 4.

1.2 SCOPE AND LIMITATIONS

Strength of metallic tension fittings.

Out of the scope of methodologies presented in this chapter:

- ✓ Tension angles made of composite materials
- ✓ Bolts loads calculation
- ✓ Configurations out of those applicable ranges declared along the discussion section

2 STRENGTH OF TENSION FITTINGS

2.1 CORNER FITTING

This type of tension fitting is referred as “Angle Fitting” in Niu’s book, see Ref. 1, chapter §9.9(A). Within this document “angle” is reserved to single L-shape geometries. Method to compute the strength of this type of fitting shall be that on that reference.

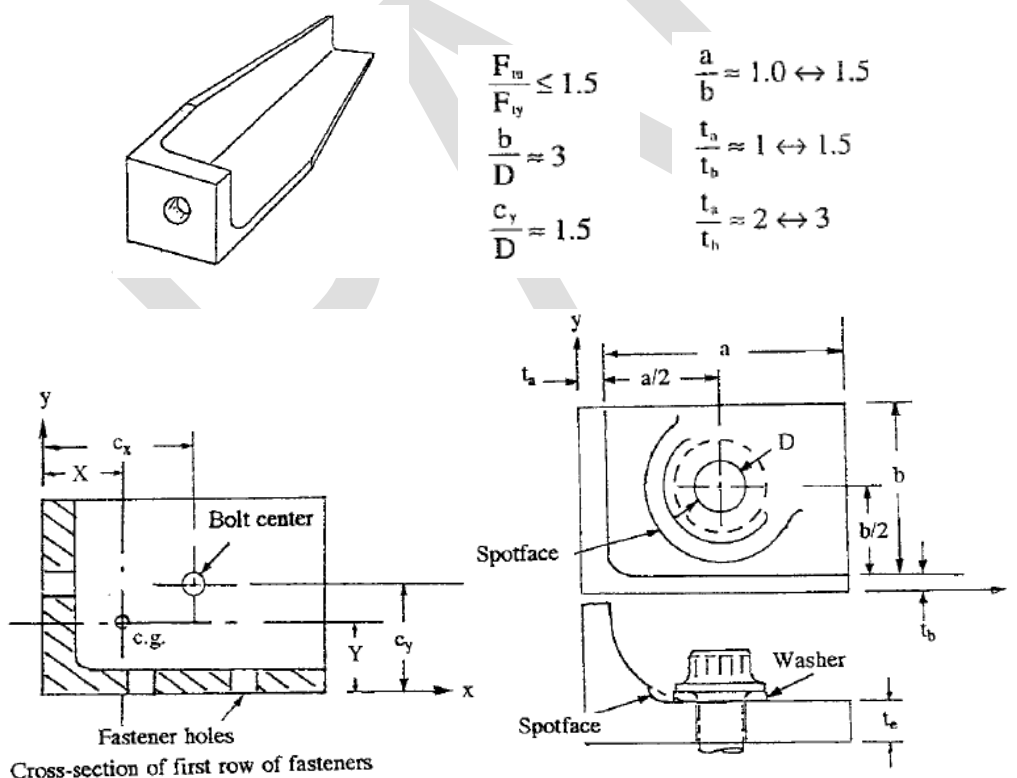


Figure 2. Corner fitting (Niu's Fig. 9.9.5)

a) Fitting wall/leg

Perform a standard beam section analysis in combined axial load plus bending moment on the L cross section (plastic bending permitted in ultimate loads)

b) Fitting end/base

- Compute the fictitious ultimate bending stress F_{bu} of the end plate of the fitting. Use numerical tools for accounting material plasticity or approximate this value per Cozzone's method in rectangular sections (Bruhn's C3.4). Typical values of the ratio F_{bu}/F_{tu} are between 1.4 and 1.5 for metal alloys used in aerospace applications.

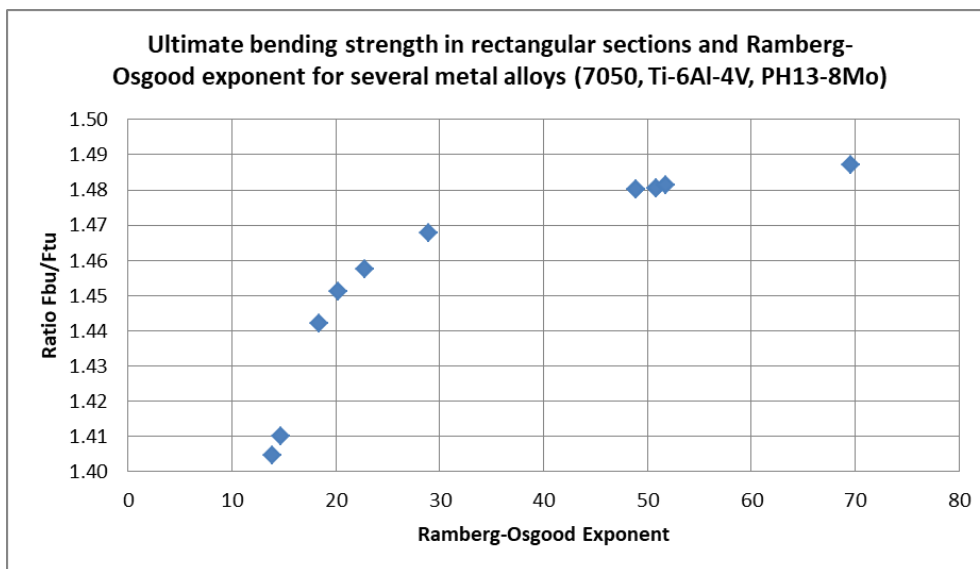


Figure 3. Ratio F_{bu}/F_{tu} for some metal alloys

- Compute the maximum bending stress on the end bolted plate from Niu's method and graph in Fig. 9.9.6 (parameter k_a) as:

$$f_{bu} = k_a \cdot P / t_e^2 \quad \dots \text{where } P \text{ is the fitting tension force, } t_e \text{ the end plate thickness}$$

- The reserve factor will be:

$$RF = F_{bu} / f_{bu}$$

c) Pull-through (shear through bolt hole)

Compare the shear stress across the cylindrical section of the end plate under the bolt head/washer (calculated as shown below), with the ultimate shear stress of the material:

$$\tau = P / (0.6 \cdot \pi \cdot D_w \cdot t_e) \quad \dots \text{where } D_w \text{ is the bolt head/washer outer diameter}$$

$$RF = F_{su} / \tau$$

For double corner fittings use method Ref. 1, chapter §9.9(C).

2.2 CHANNEL FITTING

Method for checking the strength of this type of fitting shall be similar to that explained for the corner fitting but using its particular geometry for the wall analysis, and using parameter k_c in Niu's graph Fig. 9.9.6 for computing the bending stress in the end base (Ref. 1, chapter §9.9(B)).

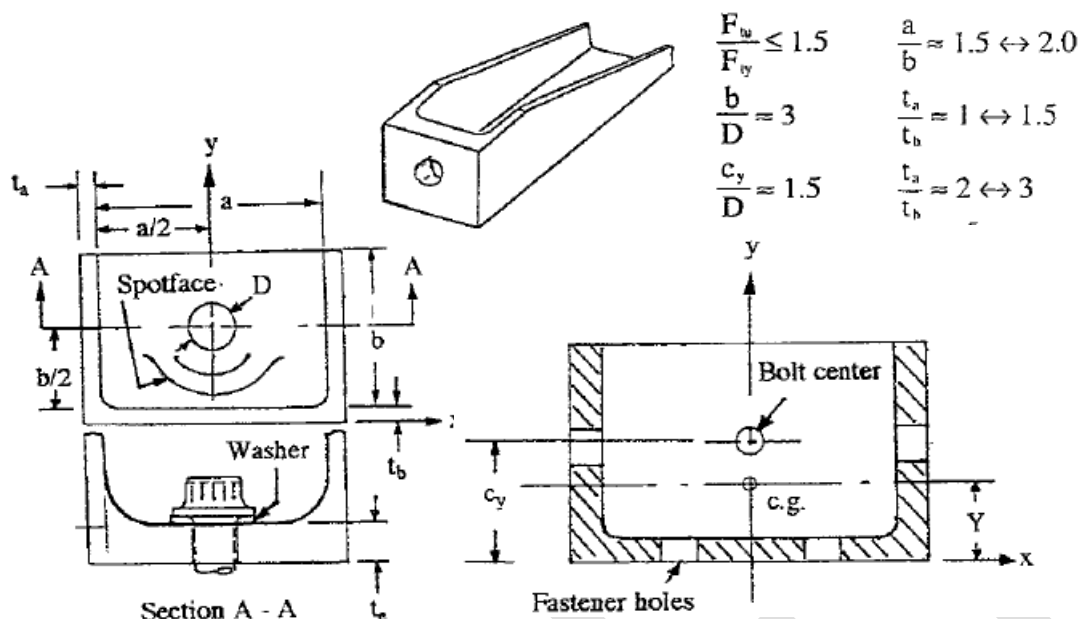


Figure 4. Channel fitting (Niu's Fig. 9.9.7)

2.3 L-SHAPED / ANGLE FITTING (RIGID LEG)

The analysis of a L-shaped angle with bolts in tension at the foot-plate and a rigid leg shall be performed assuming the bolted foot-plate behaves like a clamped-guided beam between the bolt and the leg supports, with the simplifications described in following paragraphs.

Effective widths for calculating bending at base flange

Original assumption of clamped/guided beam are considered not realistic at any section widths out of this range. Corresponds to influence lines at 45° from bolt position (see figure)(*).

- ✓ Effective width at leg: $b_l = \min[b , d + 2 \cdot (a - c - t/2)]$
- ✓ Effective width at bolt: $b_b = \min(b , 4 \cdot d)$

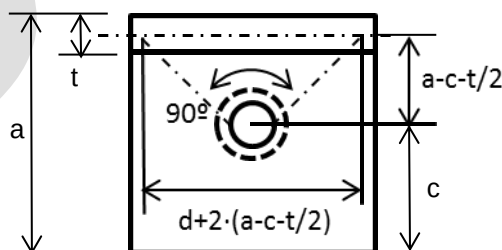


Figure 5. Geometrical considerations for the effective width at foot-plate

(*) The proposed effective field of view for bending stress calculation, equal to 90°, can be justified as a conservative simplification of the stress distribution expected for a cantilever plate under a concentrated load, eg. per Niu's Fig. 7.2.6 (for this specific case adding the bolt diameter).

Analysis of foot/base in bending

- ✓ Length of beam (lever arm in bending analysis), ultimate bending moment and stress in foot (at bolt and leg sections) can be calculated as follows:

$$L = a - c - t/2 - d/2$$

$$M_f = SF \cdot P_{LL} \cdot L / 2$$

$$\sigma_{b_lin} = 6 \cdot M_f / [\min(b_l, b_b) \cdot t^2]$$

- ✓ Reserve Factor for foot bending (UL) : $RF_{fb} = F_{bu} / (\sigma_{b_lin} \cdot FF)$
- ✓ Reserve Factor for foot bending yield at LL : $RF_{fby} = F_{ty} / [(\sigma_{b_lin} / SF) \cdot FF]$

Analysis of bolt/nut in tension

- ✓ Effective edge distance for prying effect: $c_e = \min(c, 8 \cdot t^{0.5})$...with t in mm (*)
- ✓ Ultimate tension load per bolt : $P_b = SF \cdot P_{LL} + [M_f / (c_e + d/2)]$
- ✓ Reserve Factor for bolt/nut in pure tension : $RF_{bolt} = T_{b_all} / (P_b \cdot FF)$

Analysis of foot in shear

- ✓ Ultimate shear stress around bolt : $\tau = 3 \cdot P_b / (\pi \cdot d_e \cdot t)$
(shear stress amplified at side facing leg)
- ✓ Reserve Factor for foot shear (UL) : $RF_s = F_{su} / (\tau \cdot FF)$

Analysis of leg in tension + bending

- ✓ Ultimate tension stress in leg : $\sigma_l = SF \cdot P / (b_l \cdot t)$
(the 4·d limitation for effective width is not applied within leg/web)
- ✓ Ultimate linear bending stress in leg : $\sigma_{b_lin}' = 6 \cdot M_f / (b_l \cdot t^2)$
- ✓ Reserve Factor for leg (UL) : $RF_l = 1 / [FF \cdot (\sigma_l / F_{tu} + \sigma_{b_lin}' / F_{bu})]$
- ✓ Reserve Factor for leg yield at LL : $RF_{ly} = F_{ty} / [FF \cdot (\sigma_l + \sigma_{b_lin}') / SF]$
(required as plastic bending was accounted at ultimate load level)

(*) These cut-off value for the edge distance as effective prying lever arm for bolt load calculation, equal to $8 \cdot t^{0.5}$ is (with t in mm) is taken from the legacy method described in Ref. 5.

2.4 T-FITTING

The analysis of the T-fitting (symmetrical) shall be done in a similar manner as for the L-shaped fitting, taking into account that the total load is sustained by two bolts and the web will be in pure tension (no web bending as symmetrical loading with no eccentricity).

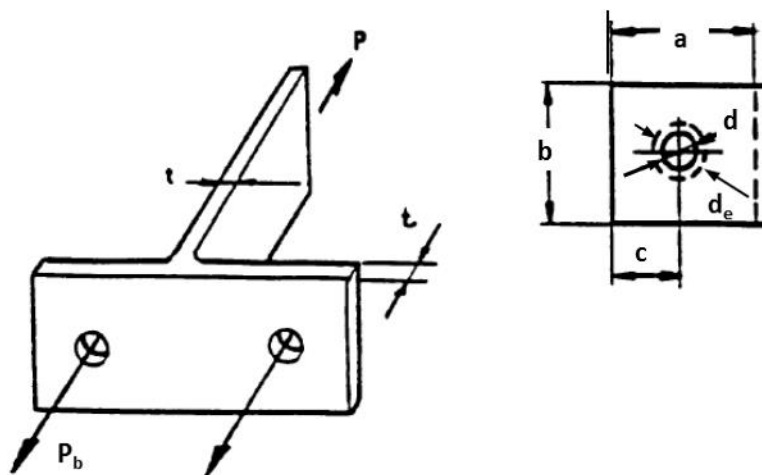


Figure 6. T-fitting general arrangement and dimensions

2.5 FOLDED CLIP & EXTRUDED ANGLES

The strength of folded sheets and extruded profiles with rivets/bolts loaded in tension shall be determined by the method described in HSB 57100-01 (Ref. 3). The method refers the strength of the flange as functions (graphs) of its thickness t , the distance from the web/leg segment to the bolt e , and the material yield stress F_{ty} . It covers L-shaped profiles but also T-fitting and back-to-back L-shaped profiles.

In addition, this HSB report proposes a simple formula for calculating the bolt tension amplified by the prying effect. However, the decision about the effective edge distance to use as lever to react the prying moment on the flange is left to an engineering judgement (if the angle is rigid or not). See following chapter for further advice.

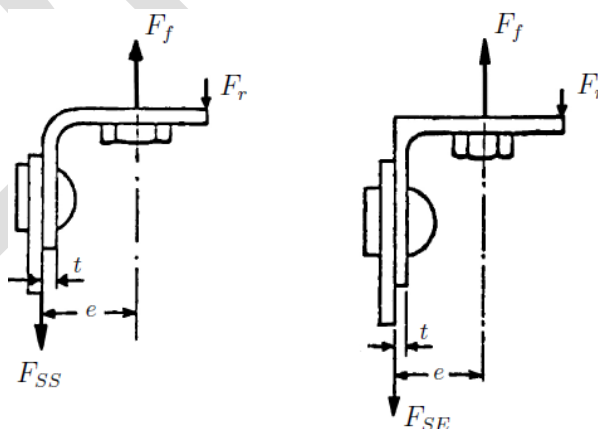


Figure 7. Folded sheet (left) and extruded profile (right) as per HSB 57100

3 PRYING EFFECT AND BOLT TENSION LOAD

For L and T shaped fittings and clips (machined, folded or extracted) the amplification of the tension load sustained by the bolt due to the prying effect shall be taking into account to perform the strength check of the bolt, nut and pull-through failure.

The calculation of the bolt tension load in these cases shall be done according the method described in sub-section 2.3 of this chapter.

4 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
a	mm	Length of foot/base
b	mm	Width of foot/base, or pitch of row of rivets parallel to leg
c	mm	Edge distance from bolt centre
d	mm	Bolt nominal (shank) diameter
d _e	mm	External diameter of bolt-head or nut
e	mm	Distance from bolt centre to leg/wall plate
FF	-	Fitting Factor
F _{su}	MPa	Ultimate shear stress of the material
F _{ty}	MPa	Yield tensile stress of the material
F _{tu}	MPa	Ultimate tensile stress of the material
F _{bu}	MPa	Fictitious ultimate bending stress of the material and cross-section (as if behaviour were linear elastic until failure)
LL	-	Limit Loads
P	N	Applied force at Limit Loads level
RF		Reserve Factor, i.e. ratio between allowable load and applied load
SF	-	Safety Factor, ratio between ultimate and limit loads, typically equal to 1.5
t	mm	Thickness
TBC		To Be Confirmed
UL	-	Ultimate Loads
σ	MPa	Axial or bending stress
τ	MPa	Shear stress

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[M. C. Y. Niu]	2 nd Ed.	AIRFRAME STRESS ANALYSIS AND SIZING
Ref. 2	[E. F. Bruhn]	1973	ANALYSIS AND DESIGN OF FLIGHT VEHICLE STRUCTURES
Ref. 3	HSB 57100-01	Is. H	Failure load of flanges under tensile load
Ref. 4	ESDU 84039	1986	Strength of angles and club-foot fittings (transmitting tensile loads)
Ref. 5	NT-00-ECE-063	Ed. A	PROCEDIMIENTO DE CÁLCULO DE ANGULARES Y HERRAJES

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>

Table 1. Software programs referenced in this chapter

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
0.3 27/07/2021	G. Baños	First issue	All pages

Table 2. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Methods (TEPAP-TL4)	G. Baños 27/07/2021	Florian Glock XX/XX/2021

Table 3. Approval signatures table for last revision of this chapter